INSTITUTE OF SCIENCE, TECHNOLOGY & ADVANCED STUDIES (VISTAS)
(Deemed to be University Estd. u/s 3 of the UGC Act, 1956)
PALLAVARAM, THALAMBUR, PERIYAPALAYAM, THIRUVANMIYUR - CHENNAI

AI-ASSISTED EV RANGE PREDICTION AND SMART CHARGING OPTIMIZATION SYSTEM

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

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PROJECT GUIDE

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PROBLEM STATEMENT

- EV drivers may not accurately know how much battery energy is required for a trip.
- Energy consumption varies with vehicle, battery capacity, distance, speed and battery health.
- Selecting suitable charging stations manually can increase charging time and cost.
- Existing systems may provide route and charging information but do not always combine energy prediction and charging optimization.
- Therefore, an intelligent system is required to predict EV range and recommend an efficient charging plan.

Need: An integrated AI-based system for EV energy prediction, SOC estimation, route planning, and reserve-constrained charging optimization.

OBJECTIVES

- Predict EV energy consumption and estimate battery SOC using Machine Learning based on vehicle, battery, route, speed and battery-health features.
- Identify suitable charging stations and optimize charging location, charging amount and charging time while maintaining a minimum 25% SOC reserve.
- Develop an interactive dashboard supporting route selection, energy prediction, charging-station recommendation, trip monitoring and charging alerts.

LITERATURE SURVEY

PAPER	YEAR	GENERAL SURVEY	ADVANTAGES	LIMITATIONS
Optimal Electrical Vehicle Charging Planning and Routing Using Real-Time Trained Energy Prediction With Physics-Based Powertrain Model	2024	Combines energy prediction with route and charging planning.	Improves route efficiency and charging decisions.	Complex LSTM/MILP framework.
Federated Learning-Based Electric Vehicle Energy Consumption Prediction and Charging Station Recommendation	2025	Predicts energy/range and recommends suitable charging stations.	Integrates prediction with station recommendation and supports distributed learning.	Federated architecture adds complexity.
Optimizing Electric Vehicle Energy Consumption Prediction Through Machine Learning and Ensemble Approaches	2025	Compares ML and ensemble methods for EV energy prediction.	Helps identify accurate prediction models.	Mainly focuses on energy prediction.
Sequence-Aware Energy Consumption Prediction for Electric Vehicles Using Pre-Trip Realistically Accessible Data	2025	Uses pre-trip information and travel sequences to predict energy consumption.	Uses information available before the trip for realistic prediction.	Mainly focuses on prediction.
Electric Vehicles Charging Station Allocation Based on Load Profile Forecasting and Dijkstra's Algorithm for Optimal Path Planning	2025	Focuses on charging-station selection and optimal path.	Improves station selection	route planning. Mainly focuses on station/path planning.

RESEARCH GAP

- Existing studies mainly focus on energy prediction, route planning, or charging-station selection separately.
- Limited integration of energy prediction, SOC estimation, route planning and charging optimization in one system.
- Complex approaches may require advanced deep-learning or physics-based models.
- The proposed system integrates these functions into a single EV trip-planning application.
- Uses XGBoost, Dijkstra-based routing and PuLP/MILP optimization.

Proposed Contribution

Integrated Chennai-focused EV trip planning with energy-aware SOC estimation and reserve-constrained charging optimization.

PROPOSED SYSTEM

AI-Based Energy Prediction
XGBoost regression predicts EV trip energy consumption from vehicle, battery, route, environmental and driving-condition features.

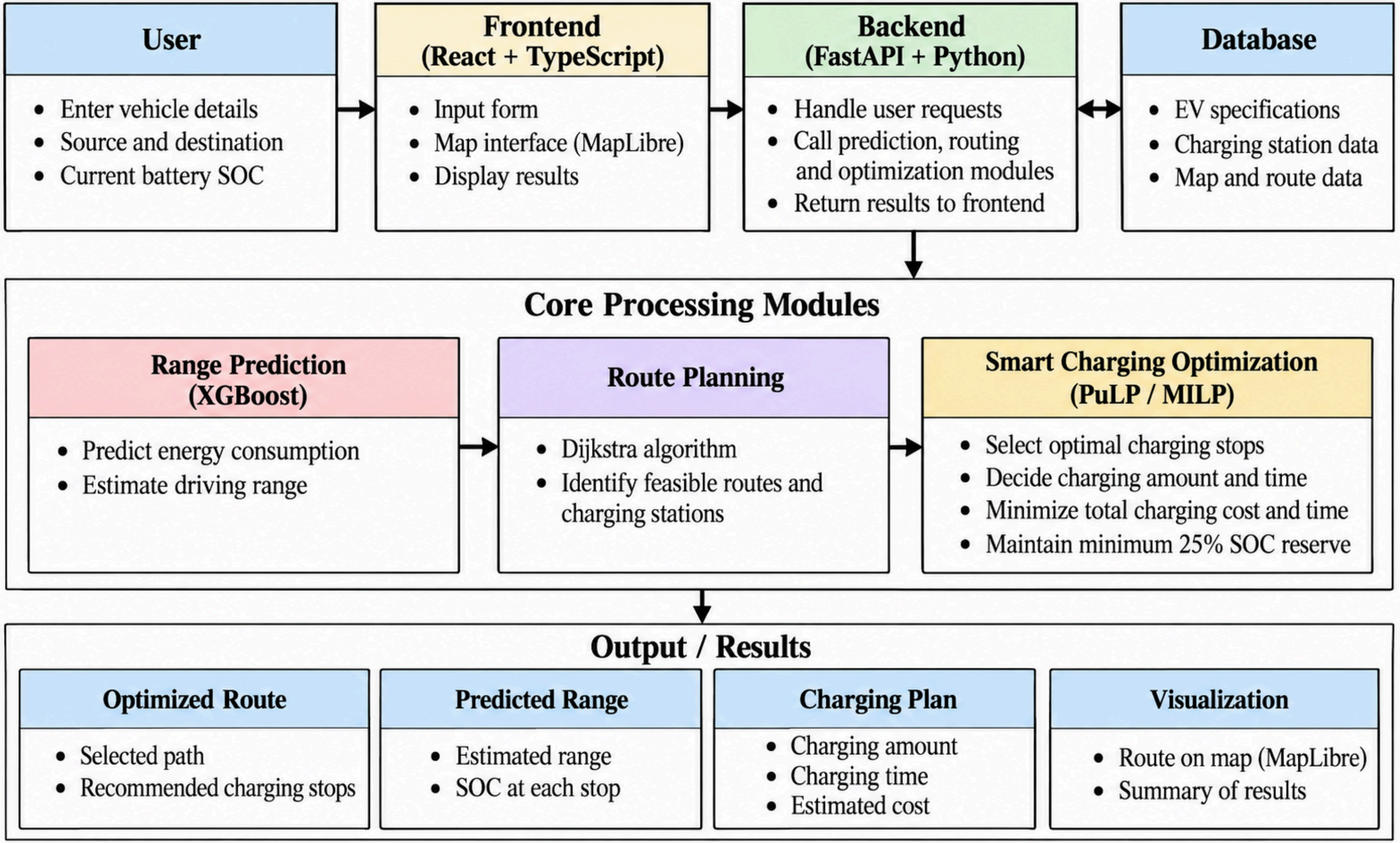
Route Planning
OSRM and Dijkstra-based processing generate and evaluate primary and alternative routes.

SOC and Range Estimation
Predicted energy is combined with initial battery SOC to estimate remaining energy, arrival SOC and safe driving range.

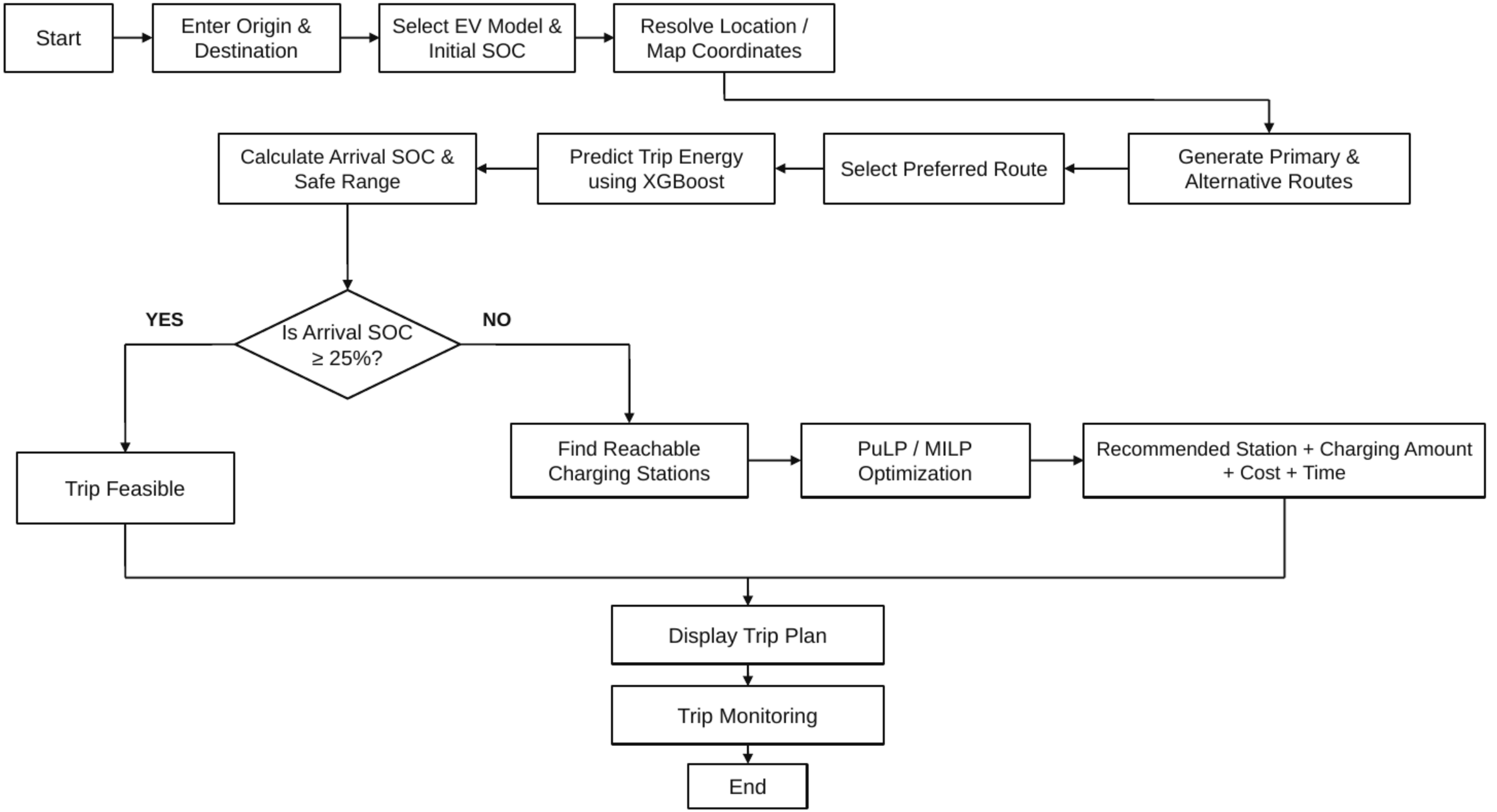
Smart Charging Optimization
Reachable charging stations are filtered and PuLP-based MILP selects charging location and charging quantity under the 25% SOC reserve constraint**.**

Interactive Dashboard
React + TypeScript + MapLibre + FastAPI dashboard displays routes, predicted energy, SOC, charging recommendations and trip-monitoring information.

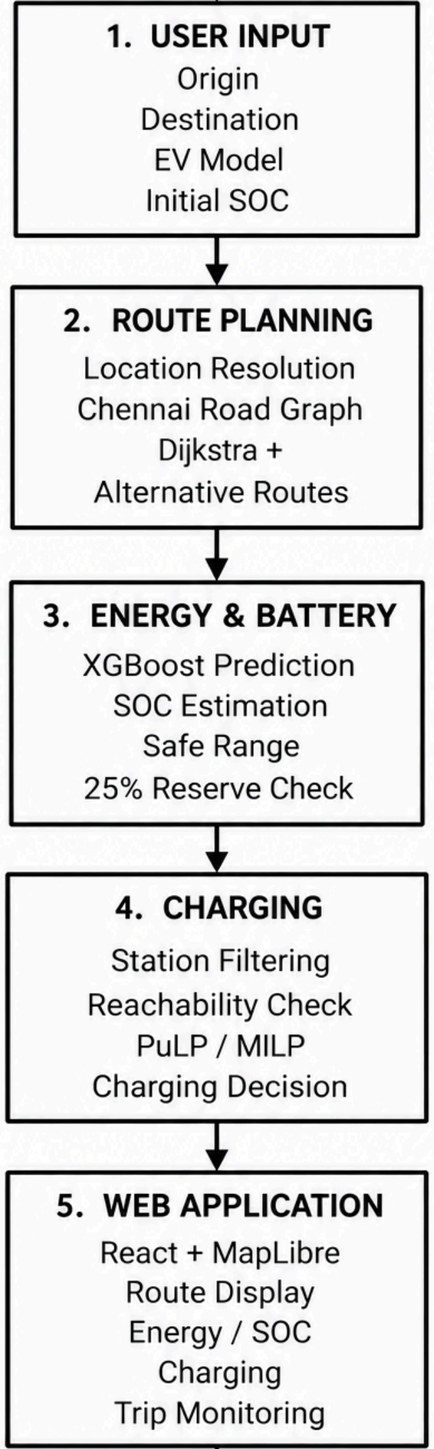
SYSTEM ARCHITECTURE



EV RANGE PREDICTION & SMART CHARGING FLOWCHART



FLOW DIAGRAM



SOFTWARE AND HARDWARE

Technology	Purpose
React + TypeScript	Frontend
MapLibre GL JS	Interactive map
FastAPI + Python	Backend / API
OSMnx + NetworkX	Road graph & routing
Dijkstra	Primary route
MongoDB	Data & geospatial storage
XGBoost	Energy prediction
PuLP / MILP	Charging optimization
GCC GIS + PIN Data	Chennai location resolution

Technology	Purpose
Processor	Intel i5 / Ryzen 5 or better
RAM	8 GB minimum
Internet	Required for routing/API features

MODULES

- Module 1 - Energy Prediction
- Module 2 - Range & SOC Estimation
- Module 3 - Charging Station Selection
- Module 4 - Smart Charging Optimization
- Module 5 – Trip Monitoring & Dashboard

MODULE 1–ENERGY PREDICTION

Dataset

- 20,000 structured EV trip records
- 30 features/columns
- 5,374 unique vehicles
- 20,000 unique trips

Vehicle-Aware Evaluation

- Training records: 16,050
- Testing records: 3,950
- Training vehicles: 4,299
- Testing vehicles: 1,075

MODELS EVALUATED

- RANDOM FOREST
- EXTRA TREES
- GRADIENT BOOSTING
- XGBOOST
- CATBOOST

XGBOOST WAS SELECTED AS THE FINAL ENERGY-PREDICTION MODEL BASED ON THE EVALUATED TEST-SET PERFORMANCE.

MODULE 2 – RANGE & SOC ESTIMATION

- Takes the initial battery SOC as input.
- Uses predicted trip energy from XGBoost.
- Calculates the remaining battery energy.
- Estimates arrival SOC and driving range.
- Checks the 25% minimum SOC reserve.
- Determines whether charging is required.

MODULE 3 – CHARGING STATION SELECTION

- Identifies charging stations near the selected route.
- Checks whether stations are reachable with the available SOC.
- Filters stations based on the 25% SOC reserve.
- Calculates distance to candidate charging stations.
- Sends feasible stations to the optimization module.

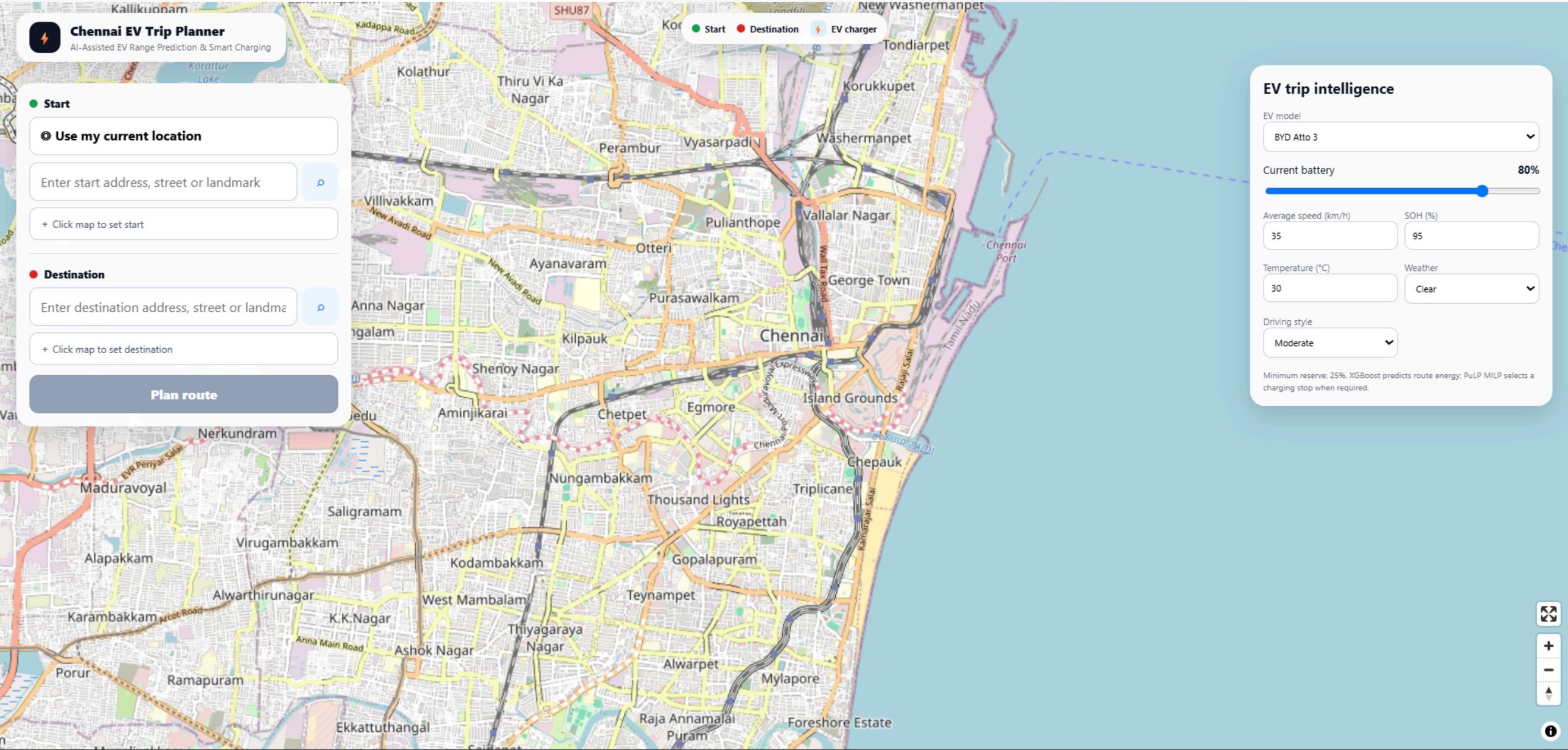
MODULE 4 – SMART CHARGING OPTIMIZATION

- Takes feasible charging stations from Module 3.
- Uses PuLP-based MILP optimization.
- Selects a suitable charging station.
- Determines the required charging amount (kWh).
- Maintains the 25% minimum SOC reserve.
- Considers charging cost and time.

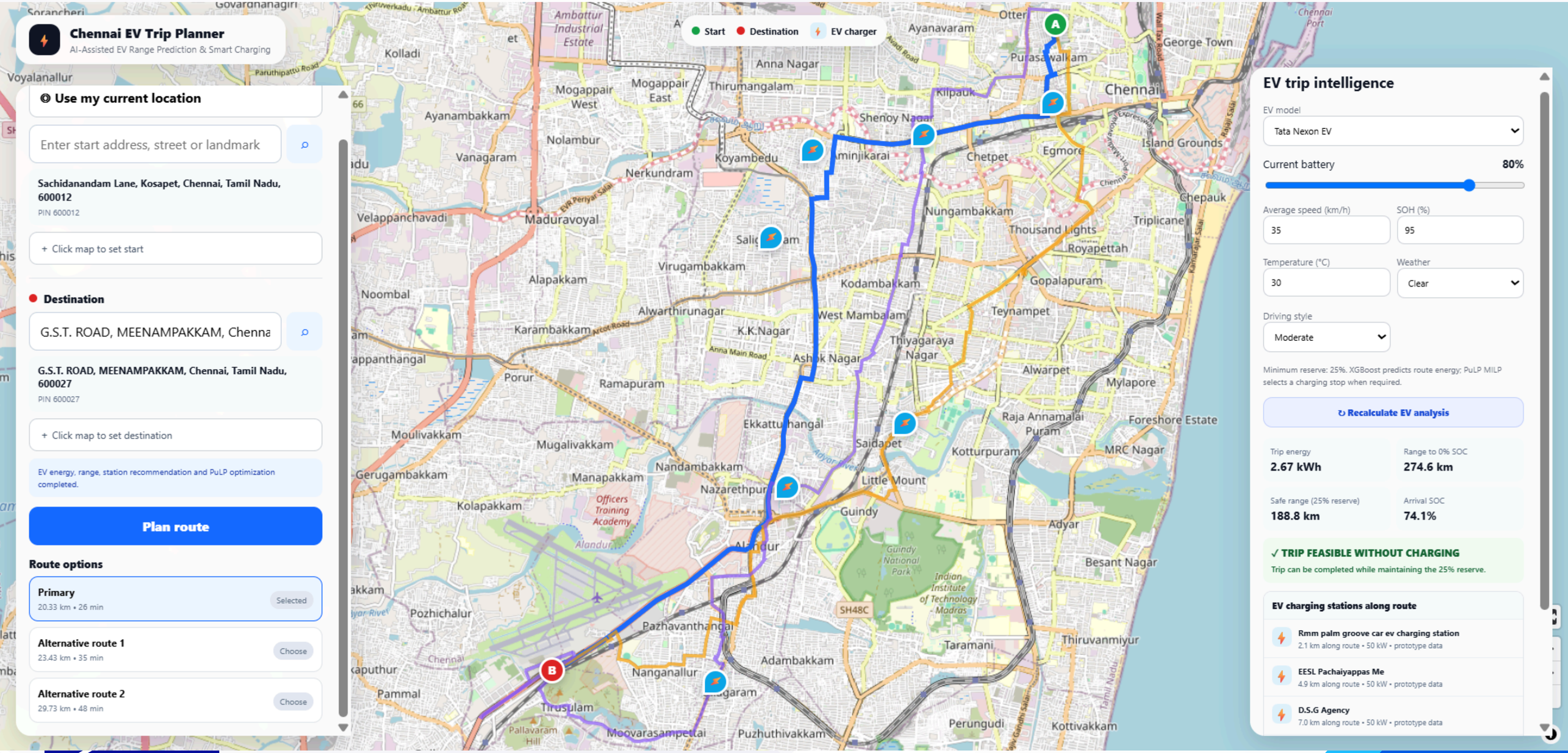
MODULE 5 — TRIP MONITORING & DASHBOARD

- Displays the selected EV route on the map.
- Shows distance and travel time.
- Displays predicted energy consumption.
- Shows current/arrival SOC and safe range.
- Displays charging station recommendations.
- Provides the final trip plan to the user.

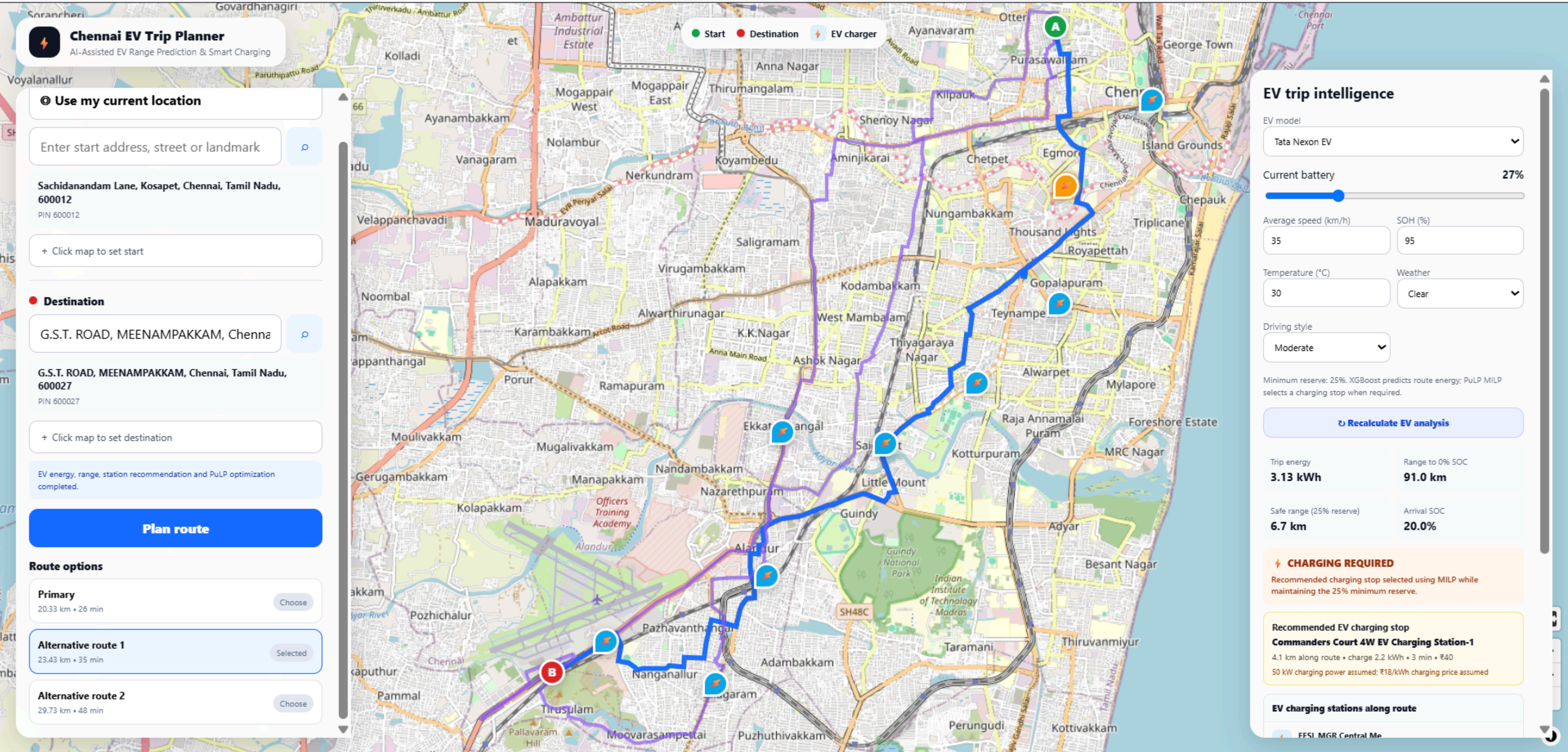
INTERACTIVE EV TRIP PLANNING DASHBOARD



ROUTE SELECTION AND ENERGY PREDICTION AND SOC ESTIMATION



TRIP MONITORING AND CHARGING ALERT



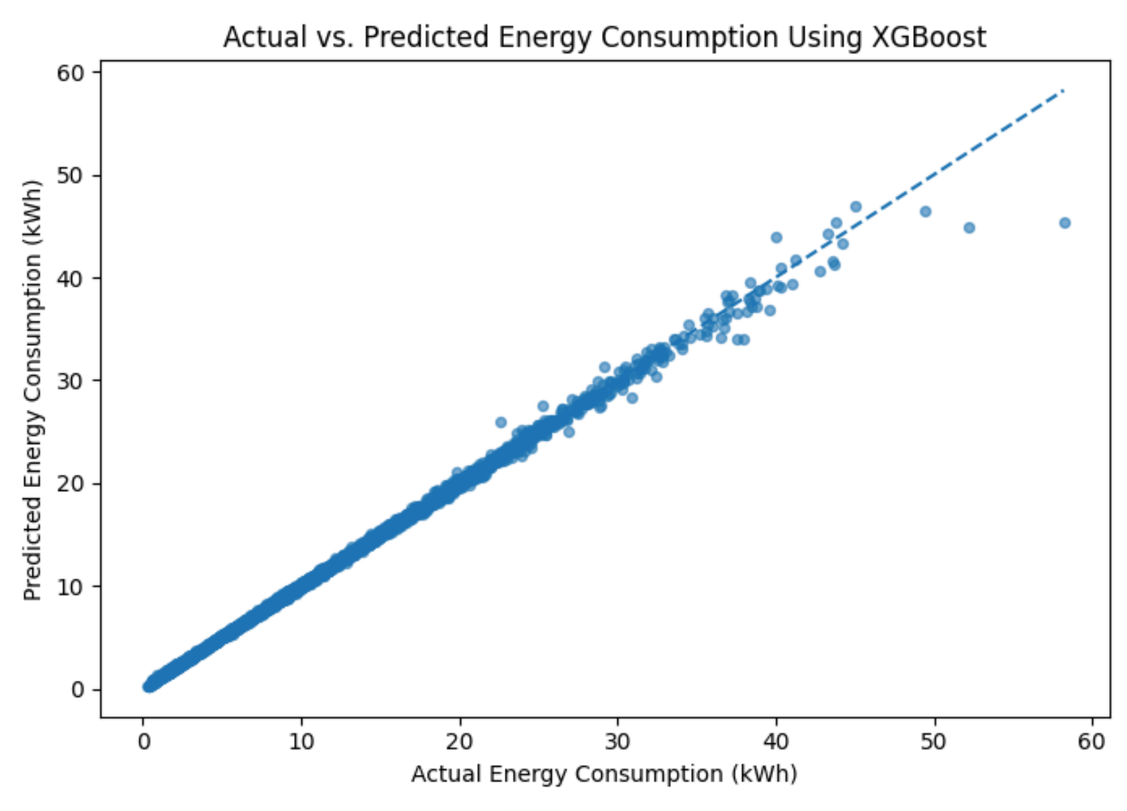
MODEL AND PERFORMANCE COMPARISON

Model	R^2	MAE	RMSE
random forest	0.9947	0.3019	0.5605
Extra tree	0.9975	0.1930	0.3838
Gradient boosting	0.9970	0.2649	0.4243
XGBoost	0.9978	0.1481	0.3613
CatBoost	0.9972	0.1744	0.4044

ACTUAL VS PREDICTED ENERGY CONSUMPTION

XGBOOST PERFORMANCE

- R^2: 0.9978
- MAE: 0.1481
- RMSE: 0.3613
- TEST RECORDS: 3,950



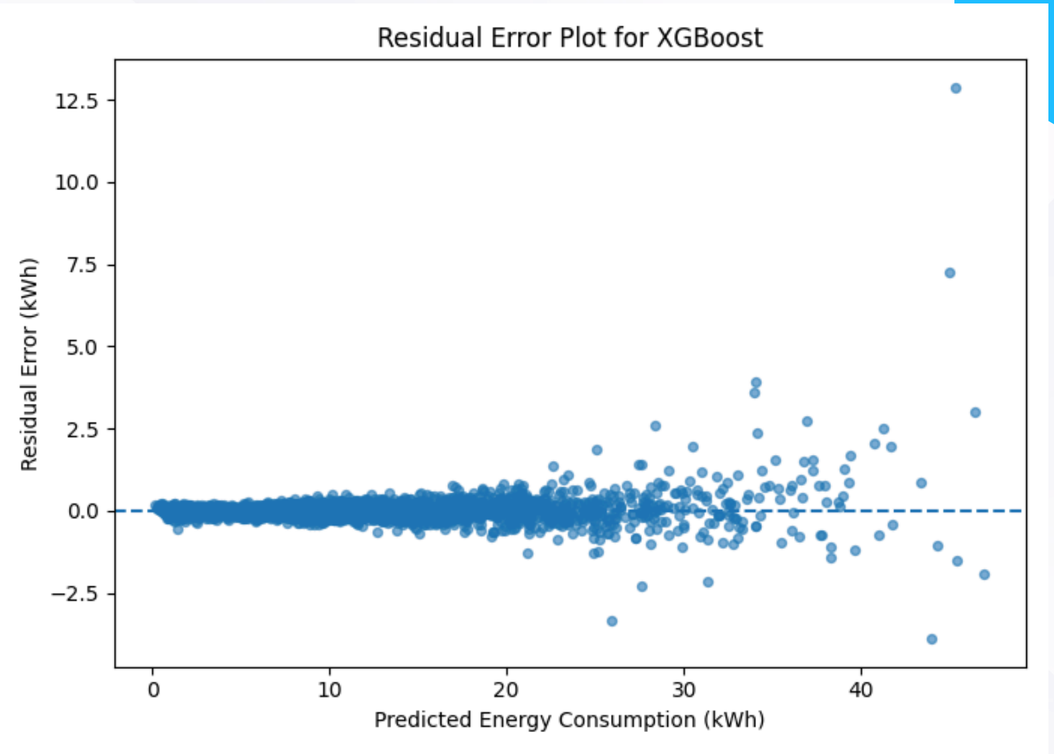
OBSERVATION :

Predicted energy consumption closely follows the actual values, indicating strong prediction performance.

RESIDUAL ERROR ANALYSIS

KEY OBSERVATION

- Most residuals are concentrated around zero error.
- Indicates small prediction errors for most test samples.
- Larger deviations occur for a small number of higher-energy trips.
- Overall error is consistent with the low MAE and RMSE.



SMART CHARGING OPTIMIZATION RESULTS

Scenario	Initial SOC	Trip Energy	Arrival SOC	Decision
S1	28%	3.51 kWh	22.2%	CHARGING
S2	20%	3.51 kWh	14.2%	NO FEASIBLE STATION
S3	20%	3.84 kWh	13.7%	NO FEASIBLE STATION

SCENARIO S1 – RECOMMENDED CHARGING

- CHARGING STATION: EESL PACHAIYAPPAS
- CHARGING AMOUNT: 1.7 KWH
- CHARGING TIME: 2 MIN
- ESTIMATED COST: ₹31

SOC RESERVE CONSTRAINT: 25%

END TO END SYSTEM TEST

PARAMETER	RESULT
EV Model	BYD Atto 3
Initial SOC	80%
Distance	14.77 km
Duration	22 min
Predicted Energy	2.55 kWh
Safe Range	192.9 km
Arrival SOC	75.8%
Charging Required	No

FINAL RESULT:

TRIP IS FEASIBLE WITHOUT CHARGING BECAUSE THE PREDICTED ARRIVAL SOC REMAINS ABOVE THE 25% RESERVE THRESHOLD.

RESULTS AND DISCUSSION

- A vehicle-aware evaluation was performed using 20,000 EV trip records.
- Five Machine Learning models were compared: Random Forest, Extra Trees, Gradient Boosting, XGBoost and CatBoost.
- XGBoost achieved the best test performance with $R^2 = 0.9978$
- The predicted energy consumption is used to estimate SOC and determine whether the trip can maintain the 25% minimum reserve.
- The dashboard provides route selection, charging-station recommendations and trip monitoring.

Model	R^2	MAE	RMSE
random forest	0.9947	0.3019	0.5605
Extra forest	0.9975	0.1930	0.3838
Gradient boosting	0.9970	0.2649	0.4243
XGBoost	0.9978	0.1481	0.3613
CatBoost	0.9972	0.1744	0.4044

Final Model: XGBoost

LIMITATIONS

- Dataset is based on structured EV trip records.
- Independent real-world fleet telemetry is not available.
- Charging cost and power may use prototype assumptions.
- Charging-station information depends on available routing/API data.
- A fixed 25% SOC reserve is currently used.
- Real-world deployment requires live vehicle and charger data.

CHALLENGES FACED

1. Data Availability

Finding suitable EV trip and charging-station data for the system.

Solution: Used available EV trip data and combined it with charging-station information.

2. Energy Prediction

Different vehicles and driving conditions affect energy consumption.

Solution: Tested multiple ML models and selected a suitable model for prediction.

3. Route Planning

Different routes can have different distances and travel times.

Solution: Added route selection so the user can choose a preferred route.

4. Charging Decisions

A trip may require charging depending on the vehicle's battery level.

Solution: Added SOC-based charging recommendations with a minimum safety reserve.

5. Trip Monitoring

Battery requirements can change while travelling.

Solution: Added trip monitoring with position and SOC updates.

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Gantt Chart

AI-Assisted EV Range Prediction and Smart Charging Optimization System

S.No	Activity	Start Date	End Date	August 2026				September 2026				October 2026			
				04-10	11-17	18-24	25-31	01-07	08-14	15-21	22-28	29-05	06-12	13-19	20-26
1	Project Topic & Scope Finalization	04 Aug 2026	08 Aug 2026												
2	Literature Survey	09 Aug 2026	15 Aug 2026												
3	Dataset Collection & Preparation	16 Aug 2026	22 Aug 2026												
4	Data Preprocessing & Analysis	23 Aug 2026	29 Aug 2026												
5	ML Model Development (XGBoost)	30 Aug 2026	05 Sep 2026												
6	ML Model Evaluation	06 Sep 2026	10 Sep 2026												
7	Route Planning Module (OSRM)	03 Sep 2026	10 Sep 2026												
8	Charging Station Module	06 Sep 2026	12 Sep 2026												
9	Charging Optimization (PuLP/MILP)	08 Sep 2026	14 Sep 2026												
10	Dashboard Development (React + MapLibre)	10 Sep 2026	16 Sep 2026												
11	System Integration & Testing	17 Sep 2026	24 Sep 2026												
12	Final Documentation & Review 3 Preparation	25 Sep 2026	15 Oct 2026												

Completed **In Progress** **Upcoming**

Project Start Date: 04 August 2026

Project End Date: 26 October 2026

THANK YOU